

Nonreciprocal Cancer–Immune Dynamics with Allee Effects and Immunological Control

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We study a spatial cancer–immune reaction–diffusion model with a cooperative strong Allee effect, nonreciprocal couplings, and adaptive immunological control. The four strong-Allee steady-state scenarios analyzed here have linearly unstable homogeneous reference states, so the steady states organize spatial routes as unstable references rather than stable attractors. In two dimensions, initially disordered fields self-organize into mesoscopic domains whose correlation lengths grow close to a diffusion-limited $t^{1/2}$ law. The main distinction is between two reciprocity regimes. For $\mathcal{R} = 0$, switching on the adaptive control $u(\mathbf{x}, t)$ strongly changes the effective driving forces: the cancer potential becomes strongly negative, while the healthy-tissue and immune potentials become positive, and the correlation structure is correspondingly reorganized. For $\mathcal{R} = 1$, the controlled and uncontrolled trajectories are much closer because the reciprocal-completion terms already impose a strong morphological bias; in this regime u acts as a secondary correction rather than the dominant organizer. We connect these correlation signatures with a thermodynamic diagnostic based on the positive diffusive entropy-production density $\sigma^+ = \sum_a D_a |\nabla \phi_a|^2$ and its integral $\Sigma_{\text{diss}}(t)$. This observable quantifies gradient relaxation and the diffusive cost of interfaces, while species-resolved contributions identify whether cancer, healthy tissue, or immune structure dominates the organization. Effective potentials and their gradients provide complementary observables for comparing the scenarios. In the reciprocal completion of the dynamics, equivalent to a free-energy-minimizing gradient flow, the Hill control has no comparable effect on diffusive entropy production; its thermodynamic signature is therefore specific to the nonreciprocal regime. The simulations are reported in nondimensional model units on a domain $[0, 8]^2$; physical lengths must be assigned by calibration to a histological, cellular, or tissue scale. Together, the correlation lengths, effective potentials, and diffusive entropy-production measures show that nonreciprocity and adaptive control select spatial scales and coupling structures rather than producing relaxation to a homogeneous equilibrium.

I. INTRODUCTION

Tumor–immune dynamics are open, driven, and spatially heterogeneous. From a nonequilibrium-physics perspective, this makes tumor progression less a question of convergence to a single homogeneous equilibrium and more a question of which transient routes, interfaces, and spatial scales are selected by the local interactions. Spatial reaction–diffusion models of tumor growth and treatment provide a natural language for this problem [1–3]: local growth and competition determine the kinetic flow, while diffusion controls the relaxation of spatial gradients and the cost of interfaces.

Classical reciprocal Lotka–Volterra or logistic descriptions are useful baselines, but they do not fully capture two mechanisms that are relevant for cancer–immune competition. First, cooperative tumor growth can generate an Allee-type response, where growth depends nonlinearly on local tumor density and may introduce sharp viability barriers [4, 5]. Second, the action of cancer on immune or healthy tissue is not generally the mirror image of the reverse action: immune cytotoxicity, tissue damage, and immune recruitment need not be pairwise reciprocal [6–8]. Such nonreciprocity can introduce rotational components in phase space, reorganize nullclines, and bias spatial routes even when homogeneous reference states are linearly unstable.

Here we study a three-field cancer–healthy–immune reaction–diffusion model with a cooperative strong Allee effect, a binary reciprocity parameter $\mathcal{R}$, and a spatially targeted Hill immunological control field. The case $\mathcal{R} = 0$ is treated as the nonreciprocal baseline, while $\mathcal{R} = 1$ represents a reciprocal-completion setting that is closer to a free-energy-gradient structure. We compare the four strong-Allee scenarios obtained by switching $\mathcal{R}$ and the Hill control on or off. The analysis focuses on three observables: spatial correlation lengths, effective population potentials, and diffusive entropy production. Together these quantities show whether control changes the geometry of tumor–immune organization or merely produces a secondary correction to a route already selected by reciprocity structure.

II. MODEL

A. Population variables

- $c(t)$ or $c(\mathbf{x}, t)$: cancer cell density.

- $s(t)$ or $s(\mathbf{x}, t)$: healthy tissue density.
- $i(t)$ or $i(\mathbf{x}, t)$: immune effector density.

B. Variational analogy and nonreciprocal transport

The model is motivated by a field-theoretic analogy with coupled population fields and gradient penalties, but the implemented reaction terms need not be globally integrable as the relaxation of a single free-energy functional. This distinction matters because the local reaction Jacobian can contain an antisymmetric component associated with nonreciprocal cancer–healthy–immune couplings. We therefore use the variational language as an organizing analogy and the transport form as the operational nonequilibrium framework.

The corresponding effective functional template takes the form:

$$F[c, s, i] = \int [f_{\text{local}}(c, s, i) + f_{\text{gradient}}(c, s, i) + f_{\text{coupling}}(c, s, i)] d\mathbf{x}, \quad (1)$$

where local terms encode growth tendencies, gradient terms capture spatial stiffness ($\frac{D_c}{2}|\nabla c|^2 + \frac{D_s}{2}|\nabla s|^2 + \frac{D_i}{2}|\nabla i|^2$), and coupling terms summarize interactions among populations.

For $\mathcal{R} = 1$, the functional includes additional reciprocity-completion terms:

$$F_{\mathcal{R}=1}[c, s, i] = F_{\text{base}}[c, s, i] + \int \left[\frac{\alpha\mathcal{R}}{4}c^2s^2 + \frac{\beta\mathcal{R}}{4}c^2i^2 + \frac{\gamma\mathcal{R}}{2}cs^2 + \frac{\eta\mathcal{R}}{2}ci^2 \right] d\mathbf{x}, \quad (2)$$

where F_{base} contains standard terms (growth, diffusion, and base interactions). In a purely gradient system, dynamics would follow from variational derivatives:

$$\partial_t c = -\frac{\delta F}{\delta c} + D_c \nabla^2 c, \quad (3)$$

$$\partial_t s = -\frac{\delta F}{\delta s} + D_s \nabla^2 s, \quad (4)$$

$$\partial_t i = -\frac{\delta F}{\delta i} + D_i \nabla^2 i. \quad (5)$$

When $\mathcal{R} = 0$, the coupling structure is the nonreciprocal baseline. The $\mathcal{R} = 1$ case introduces reciprocity-completion terms and acts as a morphological selector that reorganizes spatial patterns without stabilizing the system. In the simulations analyzed below, this selection is quantified through correlation lengths, diffusive fluxes, and the positive diffusive entropy-production density σ^+ rather than through a claim of global free-energy relaxation.

C. Base dynamics (ODE)

We start from logistic-type growth for each population and quadratic interaction terms capturing cytotoxicity, tissue damage, and immune cooperation [6]:

$$\dot{c} = r_c c f_{\text{Allee}}(c) - c(\alpha s^2 + \beta i^2) - \mathcal{R} c(\gamma s^2 + \eta i^2), \quad (6)$$

$$\dot{s} = r_s s \left(1 - \frac{s}{k_s}\right) - \gamma c^2 s + \delta i^2 s - \frac{\mathcal{R}}{2} \alpha c^2 s, \quad (7)$$

$$\dot{i} = r_i i \left(1 - \frac{i}{k_i}\right) + \delta s^2 i - \eta c^2 i - \frac{\mathcal{R}}{2} \beta c^2 i. \quad (8)$$

These ODEs are the spatially homogeneous limit of the PDE below (no diffusion), with the same local reaction structure used in the numerical implementation.

D. Cooperative Allee effect formulation

In the spatial analysis of this manuscript we use the cooperative **strong Allee** factor

$$f_{\text{Allee}}(c) = \left(\frac{c}{a} - 1\right) \left(\frac{c-a}{k_c-a}\right), \quad (9)$$

which introduces a sharp viability threshold [4, 5]. All spatial scenarios analyzed below use $a = 0.1$ in the strong-Allee branch.

E. Nonreciprocal interactions

Suppression of immunity by cancer ($-\eta c^2 i$) differs from immune cytotoxicity ($-c \beta i^2$); damage to healthy tissue by cancer ($-\gamma c^2 s$) differs from competition against cancer ($-c \alpha s^2$). These asymmetries break reciprocity and reconfigure nullclines and bifurcations.

F. Immunological control field $u(\mathbf{x}, t)$

- Reciprocity parameter: $\mathcal{R} \in \{0, 1\}$ distinguishes the nonreciprocal baseline from the reciprocal-completion setting.
- Adaptive field (Hill): $u(\mathbf{x}, t) = u_{\max} H_{\text{act}}(c; K_c, n_c) H_{\text{inh}}(i; K_i, n_i)$, with $H_{\text{act}}(c) = \frac{c^{n_c}}{K_c^{n_c} + c^{n_c}}$ and $H_{\text{inh}}(i) = \frac{K_i^{n_i}}{K_i^{n_i} + i^{n_i}}$, to prioritize regions with high c while reducing dosage when i is already high.
- Coupling: control can enter as enhanced cytotoxicity ($\beta \rightarrow \beta + u$) or as an external field in the i equation.

G. Spatial extension (PDE)

Adding diffusion yields

$$\partial_t c = D_c \nabla^2 c + r_c c f_{\text{Allee}}(c) - c(\alpha s^2 + \beta i^2) - \mathcal{R} c(\gamma s^2 + \eta i^2), \quad (10)$$

$$\partial_t s = D_s \nabla^2 s + r_s s(1 - s) - \gamma c^2 s + \delta i^2 s - \frac{\mathcal{R}}{2} \alpha s c^2, \quad (11)$$

$$\partial_t i = D_i \nabla^2 i + r_i i(1 - i) + \delta s^2 i - \eta c^2 i - \frac{\mathcal{R}}{2} \beta c^2 i. \quad (12)$$

Diffusion acts as a mode selector; with $D = \text{diag}(D_c, D_s, D_i)$ the linear spectrum is $\lambda(k) = \text{eig}(J - Dk^2)$.

Symbol	Description	Example value
r_c, r_s, r_i	intrinsic growth rates (cancer, healthy, immune)	5.84, 13.12, 10.92
k_c, k_s, k_i	carrying capacities	1, 1, 1
a	strong-Allee threshold	0.1
$\alpha, \beta, \gamma, \delta, \eta$	nonreciprocal interaction strengths	10.22, 7.60, 0.74, 5.40, 5.08
D_c, D_s, D_i	diffusion coefficients (spatial model)	0.012, 0.022, 0.022
$\mathcal{R}$	reciprocity/completion switch	$\{0, 1\}$
$K_c, n_c, K_i, n_i, u_{\max}$	Hill control (Eq. (21); Table IV)	0.05, 2, 0.2, 2, 1.0
$u(\mathbf{x}, t)$	adaptive immunological control field	bounded by $u_{\max}$

TABLE I: Model variables and parameters for the strong-Allee spatial scenarios analyzed in the main text.

III. SPATIAL DYNAMICS

Before introducing the reaction–diffusion extension, it is useful to fix the qualitative flow geometry in the (c, s) plane of the reduced kinetic model. Figure 1 shows streamlines with red markers for the projected stationary points recovered from the local steady-state catalogs and the reduced two-dimensional roots. **Roots with $i^* < 0$ in Table II are artifacts of the two-dimensional (c, s) reduction and lie outside the physical simplex; they are shown only as geometric markers in the phase portrait, not as admissible three-dimensional states.** The origin $(0, 0)$ acts as an unstable source, while $(0, 1)$, the cluster near $(1, 0)$, and the additional off-axis roots organize their neighborhoods as saddle-like structures with stable manifolds along the coordinate axes and repulsive flow into the first quadrant. This phase-space geometry prevents robust relaxation toward a stable homogeneous equilibrium and is consistent with the linear instability $\text{Re } \lambda_{\max} > 0$ reported in the summary: the steady states act as separators of transient routes rather than attractors, motivating the spatial analysis below.

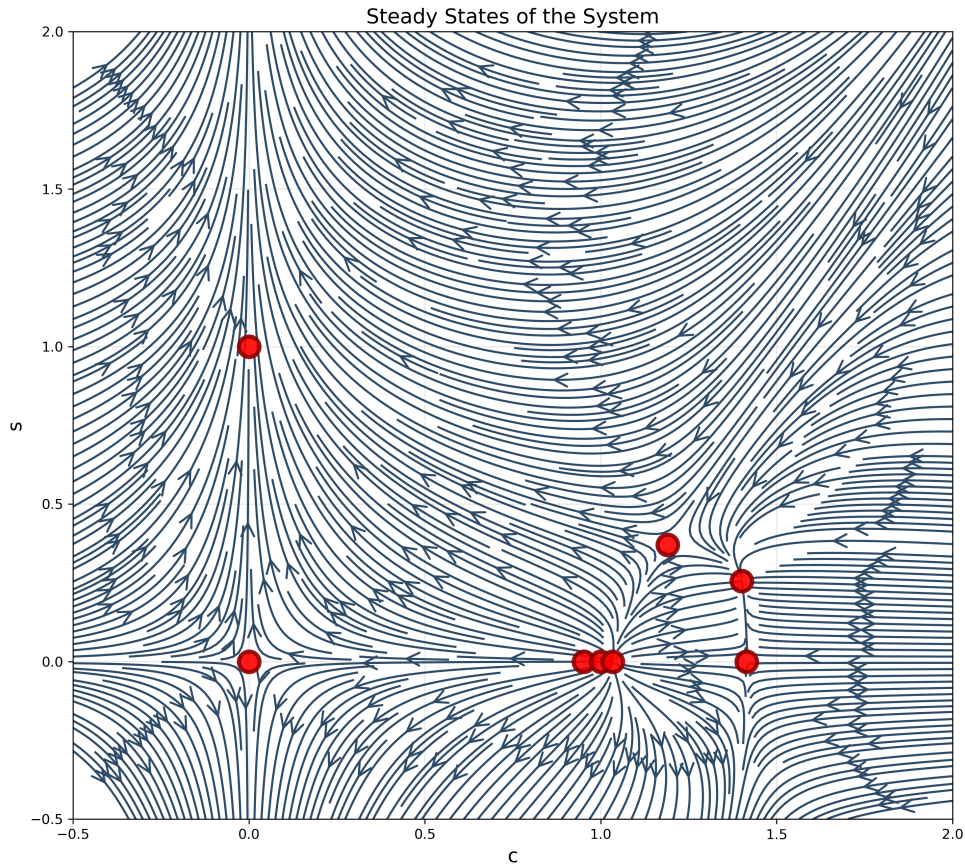


FIG. 1: Phase portrait in the (c, s) plane of the reduced two-variable kinetic model associated with the c and s dynamics. Red circles mark the projected stationary points collected from the local steady-state catalogs together with roots of the reduced two-dimensional system; points that are numerically very close in projection are rounded and merged. The streamlines show that the homogeneous references organize the flow as sources or saddle-like separators: trajectories are guided by invariant directions near the axes and by off-axis roots, but are not attracted to a global stable well. This geometry motivates the spatial simulations, where unstable local references seed pattern formation rather than homogeneous relaxation.

TABLE II: Projected stationary points marked in Fig. 1, with the corresponding immune coordinate i^* . For roots obtained from the reduced (c, s) system, i^* is reconstructed from the algebraic elimination used in the reduction; negative values indicate off-simplex reduced roots rather than physically admissible nonnegative 3D states.

c^*	s^*	i^*
0.000	0.000	1.000
0.000	1.000	0.000
0.952	0.000	-0.143
1.000	0.000	0.000
1.034	0.000	0.130
1.190	0.371	-0.083
1.401	0.256	-0.563
1.415	0.000	-0.629

The spatial model is written as a reaction–diffusion system for the population fields $\phi_a \in \{c, s, i\}$,

$$\partial_t \phi_a = D_a \nabla^2 \phi_a + R_a(c, s, i), \quad a \in \{c, s, i\}, \quad (13)$$

or, equivalently, in conservative transport form,

$$\partial_t \phi_a = -\nabla \cdot \mathbf{J}_a + R_a, \quad \mathbf{J}_a = -D_a \nabla \phi_a. \quad (14)$$

This form separates diffusive redistribution, carried by the Fickian currents $\mathbf{J}_a$, from the local nonconservative reactions R_a . For the strong-Allee scenarios used in the paper,

$$\partial_t c = D_c \nabla^2 c + r_c c f_{\text{Allee}}(c) - c(\alpha s^2 + \beta i^2) - \mathcal{R}c(\gamma s^2 + \eta i^2), \quad (15)$$

$$\partial_t s = D_s \nabla^2 s + r_s s(1 - s) - \gamma c^2 s + \delta i^2 s - \frac{\mathcal{R}}{2} \alpha s c^2, \quad (16)$$

$$\partial_t i = D_i \nabla^2 i + r_i i(1 - i) + \delta s^2 i - \eta c^2 i - \frac{\mathcal{R}}{2} \beta c^2 i. \quad (17)$$

with the cooperative strong-Allee factor

$$f_{\text{Allee}}(c) = \left(\frac{c}{a} - 1\right) \left(\frac{c - a}{1 - a}\right), \quad (18)$$

after nondimensionalizing the carrying capacity to one. The parameter $\mathcal{R}$ distinguishes the nonreciprocal baseline from the reciprocal-completion setting, while the adaptive control $u(\mathbf{x}, t)$ acts as a spatially targeted immune reinforcement defined in Sec. IV.

A. Numerical implementation

The spatial problem was solved with finite elements (FEniCSx) on a uniform 100×100 mesh over the domain $[0, 8]^2$, with final time $T = 1$ and time step $\Delta t = 10^{-3}$. Each step uses a segregated nonlinear solve (Picard inner iterations, at most three passes with tolerance 10^{-3}) coupled to PETSc SNES for the implicit field updates. Relative and absolute tolerances are $(10^{-6}, 10^{-8})$ for c and s and $(10^{-5}, 10^{-7})$ for i ; when Hill control is active, the immune update uses Newton line-search damping 0.7. Numerical stiffness is monitored operationally through diagnostic snapshots that flag solver exceptions; no eigenvalue-ratio stiffness criterion is applied in the present workflow. Table III summarizes the discretization used for all reported spatial runs; supplementary mesh/time-refinement sweeps were not performed in this revision.

TABLE III: Numerical discretization for the strong-Allee spatial scenarios reported in the main text. Convergence of $\xi(t)$, ν_{ab} , and A_{ab} with respect to mesh refinement and Δt was not quantified in additional sweeps.

Setting	Value
Domain	$[0, 8]^2$ (model units)
Mesh	uniform 100×100
Time step	$\Delta t = 10^{-3}$
Final time	$T = 1$
Realizations per scenario	$nb = 1$
FEniCSx segregated solve	Picard (≤ 3 iter., tol. 10^{-3}) + SNES

The variables are nondimensional densities and the spatial domain is reported in model units. Thus the simulations identify relative length scales, correlation growth, and interface costs. Assigning the domain to nanometers, micrometers, or millimeters requires an external calibration, for example matching $L_{\text{model}} = 8$ to a histological field of view, to an experimentally measured correlation length, or to a measured cell-migration/diffusion scale. In biological terms the patterns should be read as mesoscopic tissue organization, plausibly between cellular and histological scales, but the absolute physical unit is not fixed by the present nondimensional model. *Illustrative mapping*: if one model unit corresponds to $\sim 100 \mu\text{m}$, then $L_{\text{model}} = 8$ represents a window of $\sim 800 \mu\text{m}$, comparable to a histological field of view. This assignment is not fitted to data; experimental calibration would be required to fix the absolute scale.

Figure 3 summarizes the corresponding correlation-length diagnostics. For two fields $a, b \in \{c, s, i\}$, the normalized spatial correlation is computed from fluctuations $\delta\phi_a = \phi_a - \langle\phi_a\rangle$ as

$$C_{ab}(r, t) = \frac{\langle\delta\phi_a(\mathbf{x}, t)\delta\phi_b(\mathbf{x} + \mathbf{r}, t)\rangle_{|\mathbf{r}|=r}}{\langle\delta\phi_a^2\rangle^{1/2}\langle\delta\phi_b^2\rangle^{1/2}}, \quad (19)$$

and the characteristic length $\xi_{ab}(t)$ is extracted from the radial decay of C_{ab} . The left column reports autocorrelations, which quantify the characteristic domain size of each field; the right column reports cross-correlations, which quantify spatial co-localization between interacting populations. Across all six observables, $\xi(t)$ follows a subdiffusive growth law,

$$\xi_{ab}(t) \simeq A_{ab} t^{\nu_{ab}}, \quad \nu_{ab} \approx \frac{1}{2}, \quad (20)$$

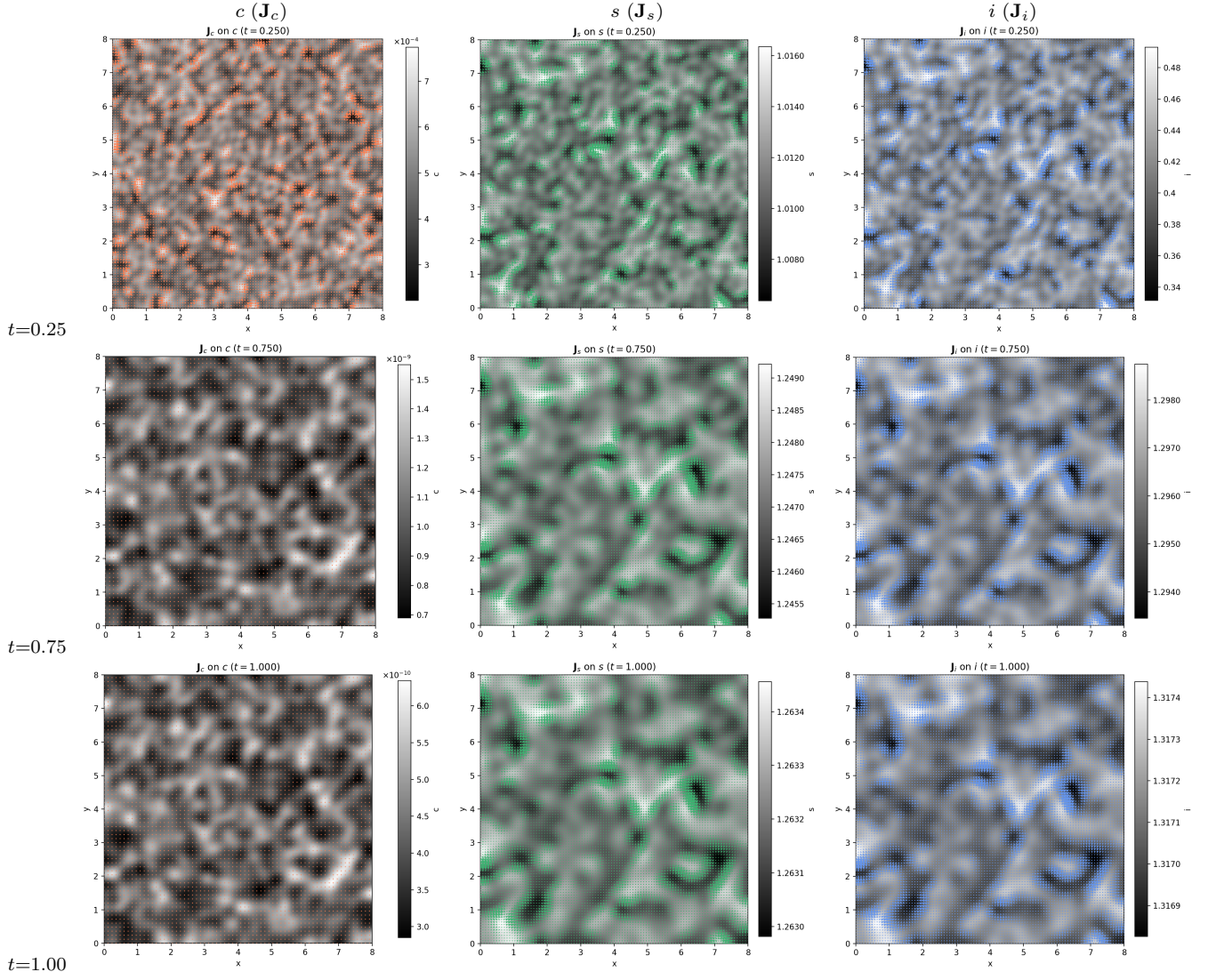


FIG. 2: Spatial evolution for the strong-Allee, reciprocal-completion setting with Hill control. Each row shows a time slice from $t = 0.25$ to $t = 1.00$; columns show the Fickian flux fields $\mathbf{J}_c$, $\mathbf{J}_s$, and $\mathbf{J}_i$ over their corresponding population densities. The initially granular texture coarsens into mesoscopic domains: cancer gradients become weak at late times, while healthy-tissue and immune fields retain aligned spatial structure, indicating coupled tissue-immune organization along common interfaces.

consistent with the classical diffusion-limited coarsening exponent [9]. The novelty in the present setting is not the exponent itself but the modulation of the prefactor A_{ab} by reciprocity structure and adaptive control. Power-law fits are reported with coefficients of determination $R^2 > 0.95$ over the fitting window used in the correlation post-processing scripts, providing an intra-series quality check (not an ensemble estimate across independent realizations). The prefactor is the physically informative quantity: adaptive control suppresses the cancer autocorrelation for $\mathcal{R} = 0$ and lowers the c - s scale, whereas $\mathcal{R} = 1$ enhances immune-centered organization, especially in i - i , c - i , and s - i .

The comparison separates two regimes. For $\mathcal{R} = 0$, the local couplings do not already impose the strong morphological bias present in the $\mathcal{R} = 1$ cases, so activating u becomes the dominant geometric intervention. It changes where gradients are generated, reduces tumor-domain coherence, and reorganizes cancer-immune co-localization. For $\mathcal{R} = 1$, the reciprocal-completion terms and the parameter set used for these scenarios already bias the dynamics toward a cancer-suppressed, healthy-immune organized route. Consequently, switching u on produces a smaller incremental change and the u off/on curves are closer in both correlations and effective-potential diagnostics. This morphological interpretation is the basis for the thermodynamic diagnostic in Sec. V: changes in correlation length should be accompanied by changes in the diffusive cost of interfaces and in the relative contributions of the c , s , and

i fields.

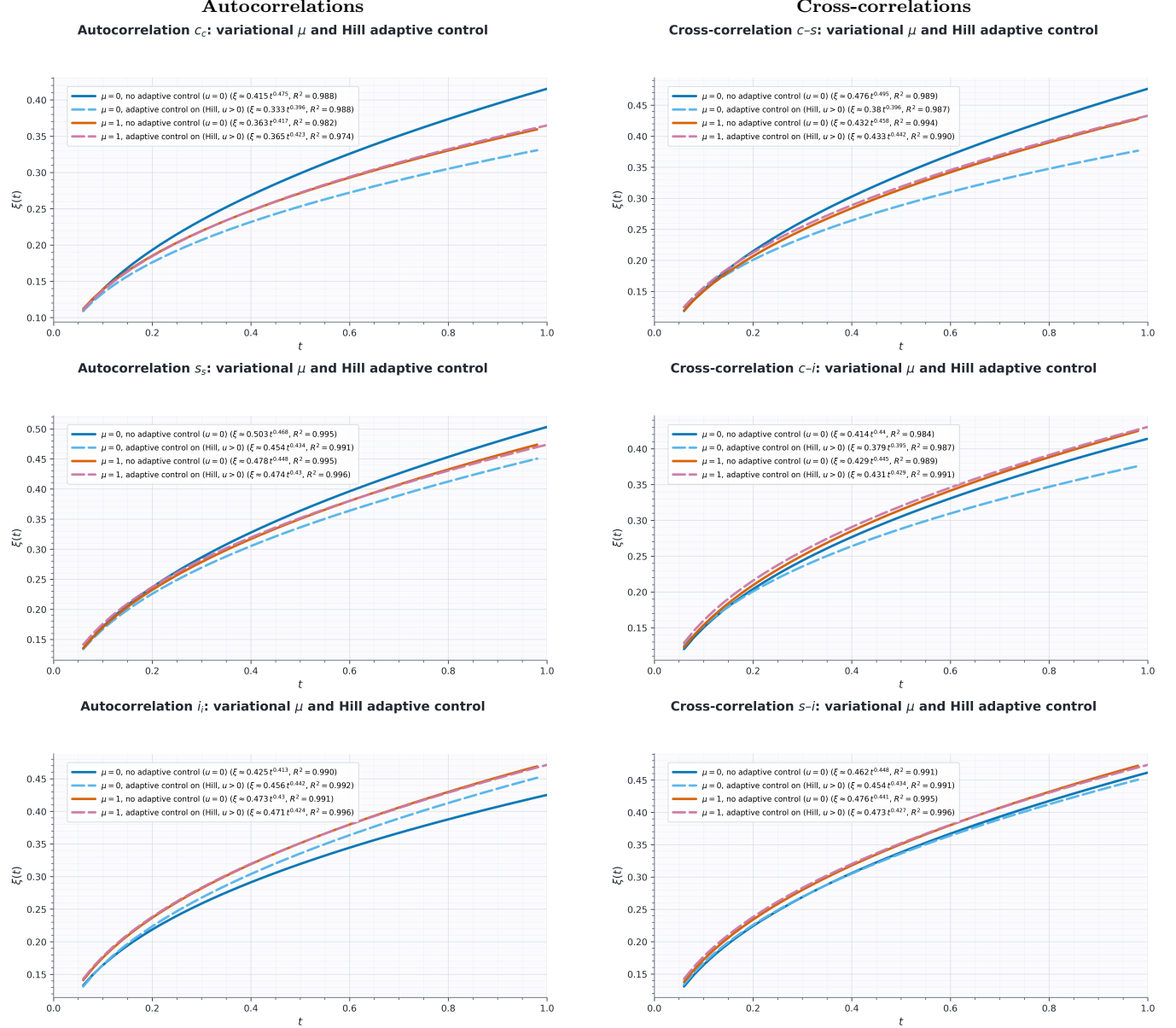


FIG. 3: Correlation-length growth in the strong-Allee spatial scenarios, arranged as autocorrelations (left column: c - c , s - s , i - i) and cross-correlations (right column: c - s , c - i , s - i). Curves compare the nonreciprocal baseline and the reciprocal-completion setting, each with and without Hill adaptive control. All observables grow with exponents close to $1/2$, consistent with diffusion-limited coarsening. The prefactors carry the mechanistic information: Hill control reduces cancer-domain coherence in the nonreciprocal baseline, whereas the reciprocal-completion setting enhances immune-centered organization and cancer-immune co-localization.

IV. IMMUNOLOGICAL CONTROL

In the spatial simulations, the active immunological reinforcement is prescribed by a Hill-type control law,

$$u(\mathbf{x}, t) = u_{\max} H_{\text{act}}(c; K_c, n_c) H_{\text{inh}}(i; K_i, n_i), \quad (21)$$

with $H_{\text{act}}(x) = x^{n_c} / (K_c^{n_c} + x^{n_c})$ and $H_{\text{inh}}(x) = K_i^{n_i} / (K_i^{n_i} + x^{n_i})$, evaluated with the positive immune density when needed. Scenarios without Hill control set $u \equiv 0$, whereas scenarios with Hill control use the parameters reported in

Table IV. This control is not a global forcing; it is spatially targeted, becoming strongest where the cancer density is high and the immune density is still insufficient.

The control should therefore be read as a field-level coupling rather than as a shift of the homogeneous fixed point. At the reference state $(c^*, s^*, i^*) = (0, 1, 0)$ the activation factor vanishes, so $u = 0$ locally and the listed eigenvalues are unchanged by turning the Hill field on. Its effect appears only after spatial fluctuations create regions with $c > 0$, where the product of activation and inhibition gates changes the local reaction route and the gradients that enter the diffusive entropy production.

Scenario	$\mathcal{R}$	Hill	u	a	r_c	r_s	r_d	α	β	δ	γ	η	K_c	n_c	K_i	n_i	$u_{\max}$	(c^*, s^*, i^*)	Re $\lambda_{\max}$
Strong-Allee, nonreciprocal baseline without Hill control	0	Off	0.1	5.84	13.12	14.0	10.22	3.0	2.0	0.74	5.08	—	—	—	—	—	—	(0, 1, 0)	16.0
Strong-Allee, nonreciprocal baseline with Hill control	0	On	0.1	5.84	13.12	14.0	10.22	3.0	2.0	0.74	5.08	0.05	2	0.2	2	1.0	—	(0, 1, 0)	16.0
Strong-Allee, reciprocal-completion setting without Hill control	1	Off	0.1	5.84	13.12	10.0	10.22	5.0	2.0	0.74	5.08	—	—	—	—	—	—	(0, 1, 0)	12.0
Strong-Allee, reciprocal-completion setting with Hill control	1	On	0.1	5.84	13.12	10.0	10.22	5.0	2.0	0.74	5.08	0.05	2	0.2	2	1.0	—	(0, 1, 0)	12.0

TABLE IV: Three-dimensional homogeneous equilibria for the strong-Allee branch used in the four spatial scenarios analyzed in the manuscript, with reaction kinetics taken from `scenarios.json`. Cases without Hill control use $u \equiv 0$, while cases with Hill control use Eq. (21). Common Hill parameters are $K_c=0.05$, $n_c=2$, $K_i=0.2$, $n_i=2$, and $u_{\max}=1$. The fixed point (c^*, s^*, i^*) and Re $\lambda_{\max}$ come from the three-dimensional fixed-point analysis with the same reaction parameters. Since the reference state has $c^* = 0$, the Hill activation vanishes there and $u=0$ locally; the control therefore acts through spatial fluctuations and interfaces rather than by shifting the homogeneous eigenvalue. Spatial diffusion coefficients are $D_c=0.012$ and $D_s=D_i=0.022$. The remaining eigenvalues in this branch are $\lambda_2 \approx -10.87$ for the nonreciprocal baseline or -11.61 for the reciprocal-completion setting, and $\lambda_3 = -13.12$.

The qualitative flow geometry near the unstable homogeneous equilibria is summarized by the phase portrait in Fig. 1, which is consistent with the projected coordinates (c^*, s^*) in Table IV. In this setting the role of control is not to create a stable homogeneous attractor. Instead, it changes where gradients and interfaces are produced, thereby modifying the spatial route by which the system approaches low-cancer regions. This is why the control effect is evaluated together with correlation lengths and diffusive entropy-production diagnostics rather than only through the local eigenvalue at the homogeneous state.

V. THERMODYNAMIC DIAGNOSTIC

The transport form introduced in Sec. III provides a direct nonequilibrium diagnostic for the spatial simulations. The diffusive currents $\mathbf{J}_a = -D_a \nabla \phi_a$ measure how gradients in c , s , and i are relaxed by transport, while the reaction terms organize the local growth, competition, immune feedback, and control response. We therefore use the gradient content of the fields as a positive diffusive entropy-production density in model units.

In nonequilibrium thermodynamics, local dissipative densities are written as sums of flux–force products. For diffusion at uniform temperature, a classical contribution has the structure

$$\sigma_S = -\frac{1}{T} \sum_a \mathbf{J}_a \cdot \nabla \mu_a + \dots, \quad (22)$$

where μ_a is a thermodynamic chemical potential. If the constitutive law is $\mathbf{J}_a = -D_a \nabla \mu_a$, then each diffusive contribution is nonnegative:

$$-\mathbf{J}_a \cdot \nabla \mu_a = D_a |\nabla \mu_a|^2 \geq 0. \quad (23)$$

The visualization and time-series post-processing used here follow the effective convention implemented in the nonequilibrium scripts: the population fields themselves are used as the transport variables driving Fickian diffusion, i.e. $\mu_a \equiv \phi_a$ at the level of the plotted diffusive fluxes. Thus the local positive density is

$$\sigma_{\text{diss},a}(\mathbf{x}, t) := D_a |\nabla \phi_a(\mathbf{x}, t)|^2, \quad (24)$$

Here σ^+ is a *metric of diffusive gradients in model units*, constructed by analogy with the De Groot–Mazur entropy-production structure, but *without formal identity* to the Onsager pair $-\mathbf{J}_a \cdot \nabla \mu_a$ when μ_a denotes a true thermodynamic chemical potential. The convention $\mu_a \equiv \phi_a$ is operational for visualization and cross-scenario comparison, not a strict thermodynamic claim. The total density plotted in the spatial σ^+ maps is

$$\sigma^+(\mathbf{x}, t) \equiv \sigma_{\text{diss,tot}}(\mathbf{x}, t) = D_c |\nabla c|^2 + D_s |\nabla s|^2 + D_i |\nabla i|^2. \quad (25)$$

The corresponding integrated time series is

$$\Sigma_{\text{diss}}(t) = \int_{\Omega} \sigma^+(\mathbf{x}, t) dA = \int_{\Omega} \sum_a D_a |\nabla \phi_a(\mathbf{x}, t)|^2 dA. \quad (26)$$

Introducing a constant effective temperature T_{eff} would only rescale this entropy-production measure by $1/T_{\text{eff}}$; it would not change the spatial patterns or the relative temporal comparisons. We therefore report Σ_{diss} directly as a diffusive organization metric: large values indicate sharp interfaces and strong transport demand, whereas decay indicates smoothing and coarsening.

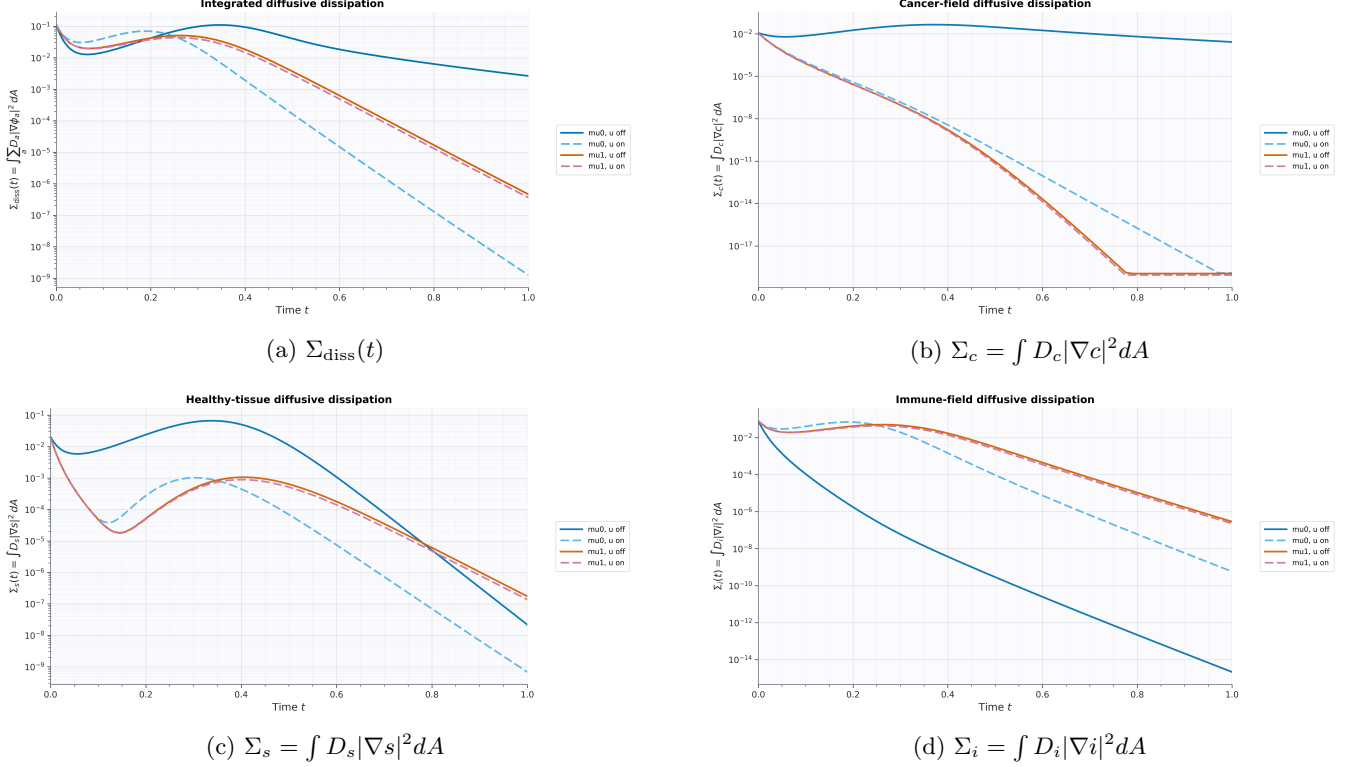


FIG. 4: Diffusive entropy-production diagnostics for the strong-Allee scenarios in the window $0 \leq t \leq 1$, sampled with $\Delta t = 10^{-3}$. The total integrated entropy production $\Sigma_{\text{diss}}(t)$ is shown together with the cancer, healthy-tissue, and immune contributions. Larger values correspond to sharper interfaces and stronger spatial gradients; decay indicates diffusive smoothing and coarsening. The uncontrolled nonreciprocal baseline retains the largest late-time gradient content, whereas Hill control strongly suppresses the cancer contribution and the reciprocal-completion curves remain close with or without control, indicating that this setting has already selected a similar spatial route.

The species-resolved panels in Fig. 4 give the temporal chronology of gradient relaxation. Early times are dominated by sharp interfaces generated from the initial heterogeneous texture. As coarsening proceeds, each compartment relaxes on its own scale, revealing whether the spatial organization is controlled by tumor fronts, healthy-tissue recovery, or immune localization.

We also compute effective potentials from the variational post-processing functional,

$$\mu_a^{\text{eff}}(\mathbf{x}, t) \sim \frac{\delta F}{\delta \phi_a}, \quad (27)$$

and use their spatial averages to compare the mean direction and intensity of the population-level driving forces. Because the reaction Jacobian can be decomposed as

$$A_{ab} := \frac{\partial R_a}{\partial \phi_b}, \quad A = S + N, \quad (28)$$

with a generally nonzero antisymmetric part N , the homogeneous reaction subsystem contains a rotational or nonreciprocal component. We therefore use the effective potentials as comparative observables for the modeled population dynamics.

This decomposition also clarifies the reciprocal reference case. If the reaction dynamics can be completed as a gradient flow,

$$R_a(\phi) = -M_a \frac{\delta F}{\delta \phi_a}, \quad M_a > 0, \quad (29)$$

then the antisymmetric part vanishes ($N = 0$) and the free-energy functional decreases monotonically,

$$\frac{dF}{dt} = - \sum_a \int_{\Omega} M_a \left| \frac{\delta F}{\delta \phi_a} \right|^2 dA - \sum_a \int_{\Omega} D_a \left| \nabla \frac{\delta F}{\delta \phi_a} \right|^2 dA \leq 0. \quad (30)$$

The nonreciprocal cases analyzed below should be read against this gradient-flow baseline: they need not minimize a single global F , and the Hill control can alter the spatial redistribution of gradients rather than simply accelerating descent along a free-energy landscape.

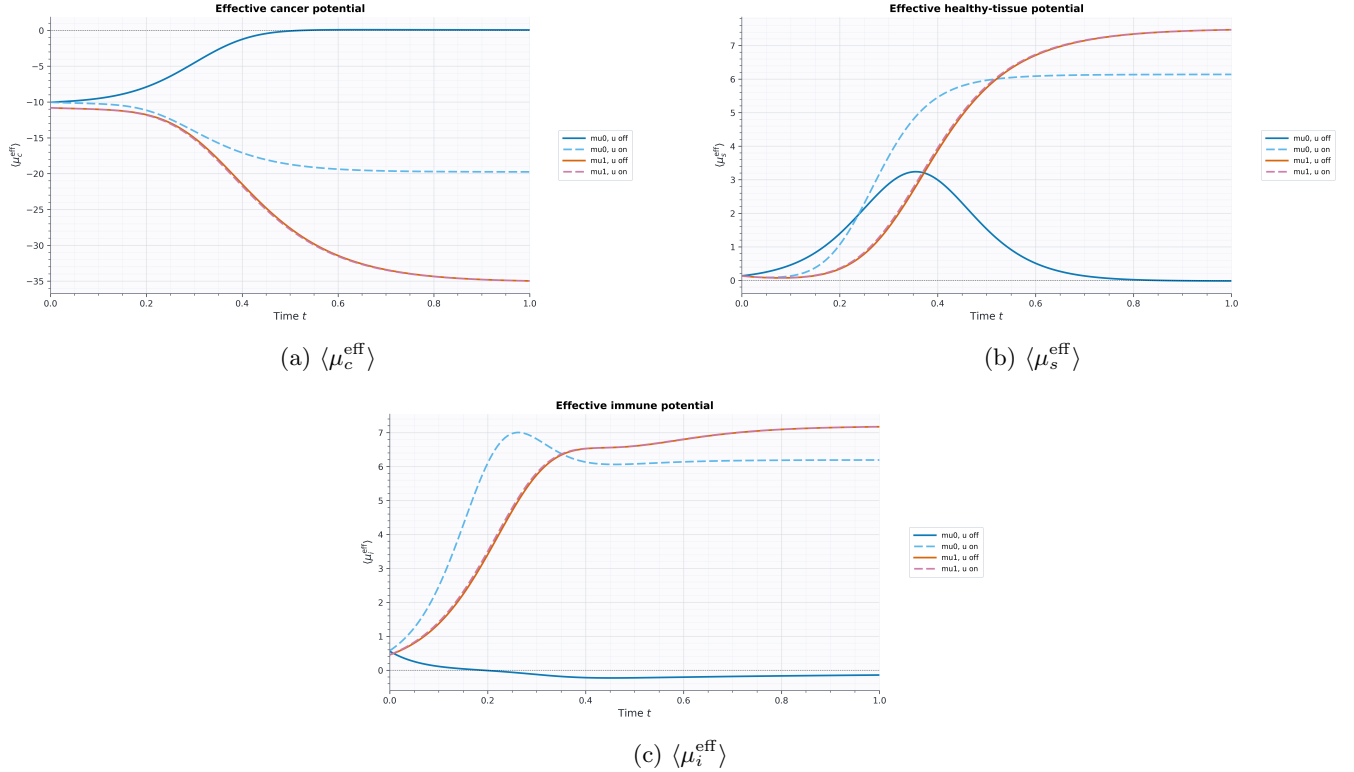


FIG. 5: Spatial averages of the effective cancer, healthy-tissue, and immune potentials for the four strong-Allee scenarios. The panel highlights how control and reciprocity structure modify the mean driving forces: in the nonreciprocal baseline, Hill control shifts the cancer potential from weakly favorable to strongly suppressive and raises the healthy-tissue and immune potentials; in the reciprocal-completion setting, controlled and uncontrolled trajectories nearly overlap, showing that the additional coupling structure already imposes the dominant thermodynamic bias.

The effective potentials make the contrast between control regimes explicit. For $\mathcal{R} = 0$, activating the adaptive control changes the final averaged potentials from a weak cancer-favorable balance, approximately $\langle \mu_c^{\text{eff}} \rangle \simeq 6.2 \times 10^{-2}$, $\langle \mu_s^{\text{eff}} \rangle \simeq -1.7 \times 10^{-2}$, and $\langle \mu_i^{\text{eff}} \rangle \simeq -1.4 \times 10^{-1}$, to a strongly cancer-suppressing and healthy-immune-favoring state, approximately $\langle \mu_c^{\text{eff}} \rangle \simeq -19.7$, $\langle \mu_s^{\text{eff}} \rangle \simeq 6.1$, and $\langle \mu_i^{\text{eff}} \rangle \simeq 6.2$. This explains why the $\mathcal{R} = 0$ correlations change substantially when u is switched on: the control reverses the effective population driving forces and changes which fields organize the interfaces.

For $\mathcal{R} = 1$, the uncontrolled and controlled trajectories are much closer. Both end with strongly negative cancer potential and positive healthy/immune potentials, approximately $\langle \mu_c^{\text{eff}} \rangle \simeq -35$, $\langle \mu_s^{\text{eff}} \rangle \simeq 7.47$, and $\langle \mu_i^{\text{eff}} \rangle \simeq 7.17$. This happens because the $\mathcal{R} = 1$ scenarios already include the reciprocal-completion terms and use the stronger cancer-immune coupling set ($\beta = 5$ instead of $\beta = 3$, with $r_d = 10$ instead of 14). In this regime, reciprocity completion

has already selected a cancer-suppressed, healthy-immune organized route, so adaptive control produces a smaller additional displacement in the effective-potential and correlation curves.

Thus, the large separation between Hill-on and Hill-off curves for $\mathcal{R} = 0$ arises because the control supplies the dominant geometric bias in a nonreciprocal background. By contrast, for $\mathcal{R} = 1$, the reciprocal-completion terms already impose a cancer-suppressing, healthy-immune organized route; Hill control therefore produces only a secondary correction in both effective potentials and diffusive entropy production.

In the reciprocal completion of the dynamics, equivalently the gradient-flow limit in Eqs. (29)–(30), the Hill control has no comparable effect on the diffusive entropy production. Its thermodynamic signature is therefore tied to the nonreciprocal regime, where the control changes the spatial redistribution of gradients rather than simply relaxing along a free-energy descent.

For completeness, we also evaluate the gradient content of these effective potentials:

$$\Sigma_\mu(t) = \int_\Omega \sum_a D_a |\nabla \mu_a^{\text{eff}}(\mathbf{x}, t)|^2 dA. \quad (31)$$

In general $\Sigma_\mu \neq \Sigma_{\text{diss}}$, unless the effective potentials are linearly related to the fields in the relevant range. The primary observable in this paper remains $\sigma^+ = \sigma_{\text{diss, tot}}$, while Σ_μ is used as a complementary measure of heterogeneity in the effective driving forces.

VI. DISCUSSION

A. Physical interpretation of results

Our results highlight an intrinsically steady-state-centered organization of cancer-immune dynamics: in the strong-Allee scenarios considered here, relevant homogeneous states act as unstable references rather than attractors. This is reflected by $\text{Re } \lambda_{\text{max}} > 0$ in the local analysis; in space, the same instability manifests as pattern selection and coarsening rather than convergence to a homogeneous equilibrium. These steady states organize spatial routes, while nonreciprocity tilts the phase flow and biases the geometry of cancer-healthy-immune tracking.

Two complementary observables make this organization explicit. First, field snapshots in temporal panels (Fig. 2) show a rapid transition from disordered initial conditions to mesoscopic domains whose morphology depends on the control protocol. Second, correlation length grids $\xi(t)$ (Fig. 3) quantify the *emerging length scale* of these domains, separating single-field coherence from inter-field co-localization. Across scenarios, $\xi(t)$ grows subdiffusively, consistent with diffusion-limited coarsening [9], and is well described by $\xi(t) \sim e^{\alpha t^{1/2}}$ over the simulated horizon. The thermodynamic diagnostic in Sec. V complements this picture through the diffusive entropy production: $\Sigma_{\text{diss}}(t)$ is largest when interfaces are sharp and decreases as the fields smooth and coarsen. Its species-resolved components identify which population field controls the remaining spatial organization.

B. Role of $\mathcal{R}$ (variational and reciprocity structure)

The parameter $\mathcal{R}$ acts as a structural switch that reorganizes the spatial spectrum: for $\mathcal{R} = 1$ patterns are typically less fragmented, with smoother interfaces and fewer high-frequency features, consistent with attenuation of short-wavelength modes. This qualitative observation is supported by correlation length grids, where $\mathcal{R}$ shifts $\xi(t)$ systematically in the strong Allee scenarios (Fig. 3). The effect is strongest in immune-centered observables: i - i , c - i , and s - i increase their prefactors when $\mathcal{R} = 1$, indicating more coherent immune domains and stronger spatial tracking of tumor and tissue structure. Importantly, $\mathcal{R}$ does not stabilize dynamics in the tested parameter set; instead it functions as a *morphological selector* that changes the geometry of transient trajectories and the characteristic scale of domains.

C. Adaptive immunological control $u(\mathbf{x}, t)$ as geometric coupling

Adaptive control introduces a spatially directed action that is stronger in regions with high c and low i . Field panels (Fig. 2) show that activating/deactivating u changes the route by which domains form and fuse. Quantitatively, c - c tracks coherence of tumor domains, while c - i tracks cancer-immune co-localization, i.e., the degree to which immunity follows tumor structure. The two-column organization in Fig. 3 separates these effects visually: autocorrelations measure domain compactness within each population, whereas cross-correlations measure the geometry of interaction

between populations. In the reciprocal completion of the dynamics, equivalent to free-energy minimization, the same control has no comparable impact on diffusive entropy production; its thermodynamic signature is therefore a genuinely nonreciprocal effect. This provides a concrete and measurable bridge between mechanistic control rules and emerging spatial organization.

D. Comparison with previous models

Relative to reciprocal Lotka–Volterra models and classical immune–tumor ODE/PDE models without Allee terms [2], the key qualitative difference in the present strong-Allee setting is that cooperative growth introduces a sharp viability threshold separating collapse from proliferation, while nonreciprocity breaks the symmetry of interaction terms and biases transient routes. In spatial configurations, this combination shifts the pattern formation landscape from purely diffusion-driven instabilities to *threshold-mediated* domain formation, where the strong-Allee barrier limits domain expansion. Control protocols then act mainly by selecting spatial scale and coupling structure, rather than creating stable homogeneous coexistence.

E. Implications and limitations

The present scenario set supports a robust qualitative conclusion: within explored parameters, the system operates near unstable steady states and the biologically relevant outcome is controlled *containment vs escape* rather than asymptotic stability. Stability may still emerge in extended sweeps (e.g., reduced r_c or improved immune recruitment via larger r_d, δ), but spatial results already show that control and Allee strength strongly constrain morphology and characteristic length scales.

Limitations follow naturally. (i) The spatial horizon is short ($T = 1$ in model units), so reported scaling and morphologies should be interpreted as transient-but-organized regimes. Table III documents the baseline mesh and time step; convergence sweeps over finer meshes or smaller Δt were not carried out in this revision. (ii) We report one realization per scenario ($nb = 1$), so ensemble variability is not quantified. The power-law fits in Sec. III use R^2 only as an intra-series quality check within each single trajectory; error bars on A_{ab} or on the final effective potentials would require independent realizations ($nb \geq 3$). (iii) Adaptive control may introduce numerical stiffness in some runs. Stiffness is diagnosed operationally when solver exceptions trigger diagnostic snapshots; the SNES/Picard tolerances listed in Sec. III should be co-designed with control smoothness to ensure robust long-time integration.

VII. CONCLUSIONS

We have analyzed a spatial cancer–healthy–immune reaction–diffusion model in which cooperative strong-Allee growth, nonreciprocal interactions, and Hill-type immunological control act together to select mesoscopic organization. The homogeneous reference states considered here are linearly unstable, so they do not serve as attracting equilibria. Instead, they organize spatial routes: initially heterogeneous fields coarsen into domains whose correlation lengths grow close to a diffusion-limited $t^{1/2}$ law over the simulated window.

The reciprocity parameter $\mathcal{R}$ is the main structural selector. In the nonreciprocal baseline ($\mathcal{R} = 0$), activating Hill control supplies the dominant geometric bias: it suppresses tumor-domain coherence, reorganizes cancer–immune co-localization, and produces large changes in the effective cancer, healthy-tissue, and immune potentials. In the reciprocal-completion setting ($\mathcal{R} = 1$), the coupling structure already selects a cancer-suppressing, healthy–immune organized route. Consequently, turning Hill control on produces only a secondary correction in both the effective potentials and the diffusive entropy production.

The thermodynamic diagnostic $\sigma^+ = \sum_a D_a |\nabla \phi_a|^2$ and its integral $\Sigma_{\text{diss}}(t)$ quantify this distinction at the level of gradients and interfaces. The control has a strong entropy-production signature only when it acts on a nonreciprocal background; in the reciprocal-completion limit, equivalent to free-energy minimization, the same control loses comparable thermodynamic relevance. Thus the principal outcome is not relaxation to a homogeneous equilibrium, but selection of spatial scales, routes, and coupling structures.

These results provide a compact framework for comparing intervention strategies through mesoscopic observables: correlation lengths, effective potentials, and diffusive entropy-production measures. Future work should extend the parameter sweeps, optimize the Hill control in space and time, test ensemble variability across initial conditions, and

calibrate the nondimensional length scale against histological or cellular data.

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